

Metrics

Metrics expose HTTP requests, order creation, pending orders, and background worker activity.

[illegible]

Prometheus

collects and stores application metrics for monitoring and analysis

localhost:9090/query?g0.expr=orderflow_orders_created_total&g0.show_tree=0&g0.tab=table&g0.range_input=1h&g0.res_type=auto&g0.res_density=m...
For quick access, place your bookmarks here on the bookmarks bar. [Import bookmarks now](#)

Prometheus Alerts Status

>_ orderflow_orders_created_total Execute

Table Graph Explain

< Evaluation time > Load time: 19ms Result series: 1

orderflow_orders_created_total{instance="localhost:7114", job="orderflow", otel_scope_name="Orderflow", otel_scope_version="1.0.0"}	6
---	---

>_ orderflow_orders_pending Execute

Table Graph Explain

< Evaluation time > Load time: 11ms Result series: 1

orderflow_orders_pending{instance="localhost:7114", job="orderflow", otel_scope_name="Orderflow", otel_scope_version="1.0.0"}	0
---	---

>_ orderflow_worker_executions_total Execute

Table Graph Explain

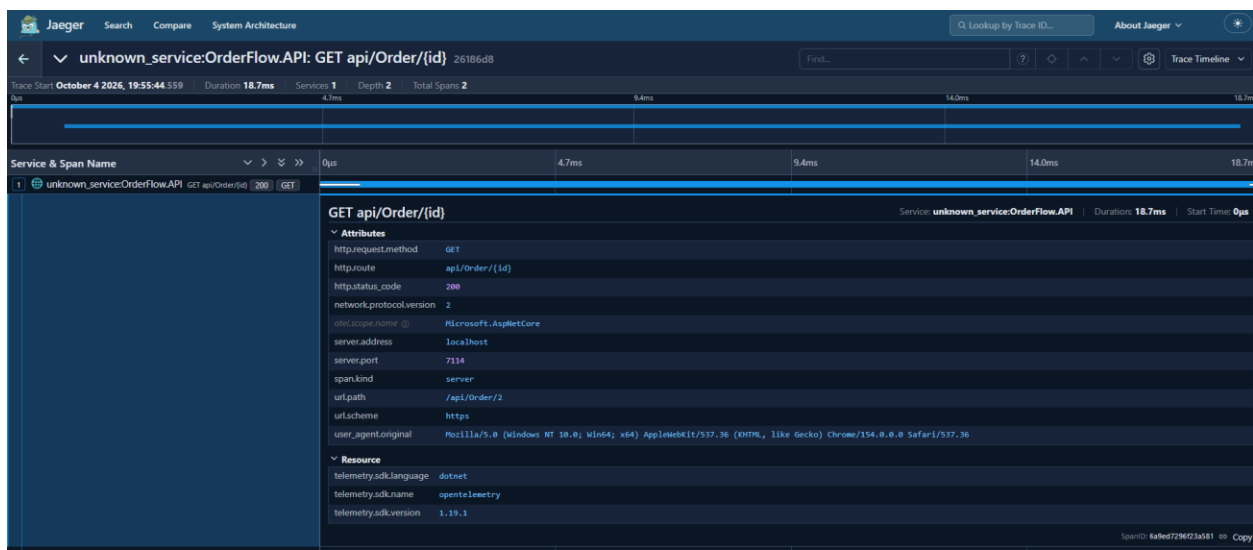
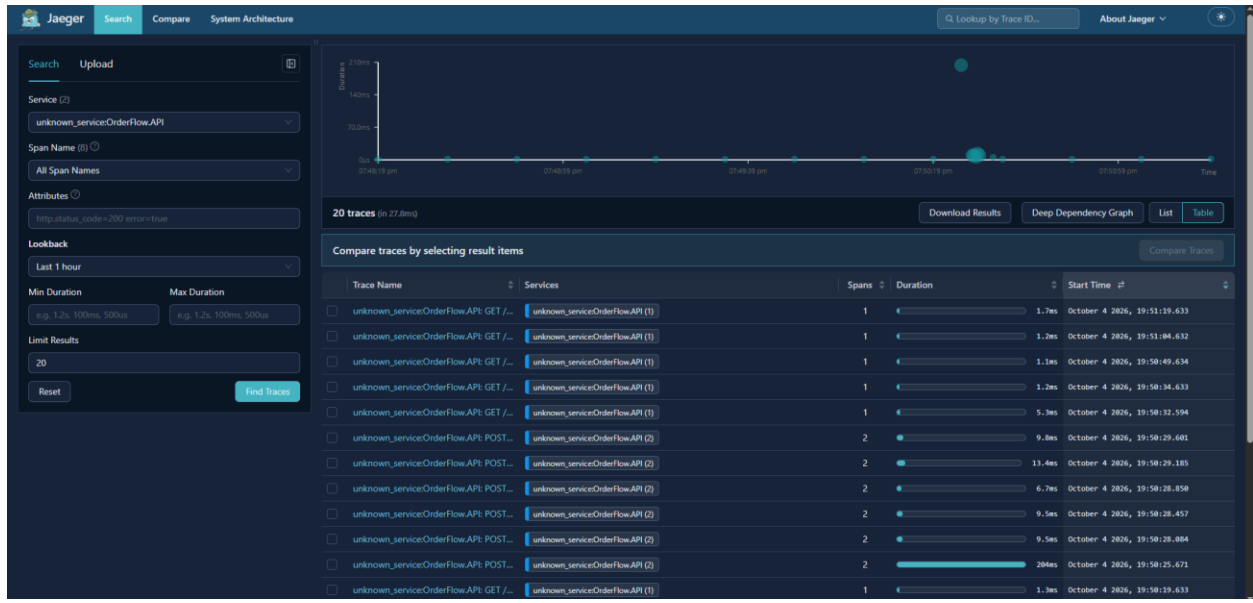
< Evaluation time > Load time: 8ms Result series: 1

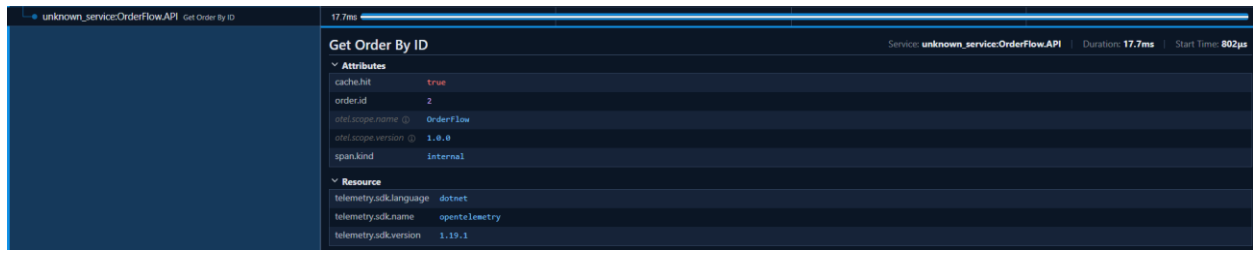
orderflow_worker_executions_total{instance="localhost:7114", job="orderflow", otel_scope_name="Orderflow", otel_scope_version="1.0.0"}	56
--	----

+ Add query

Jaeger Trace

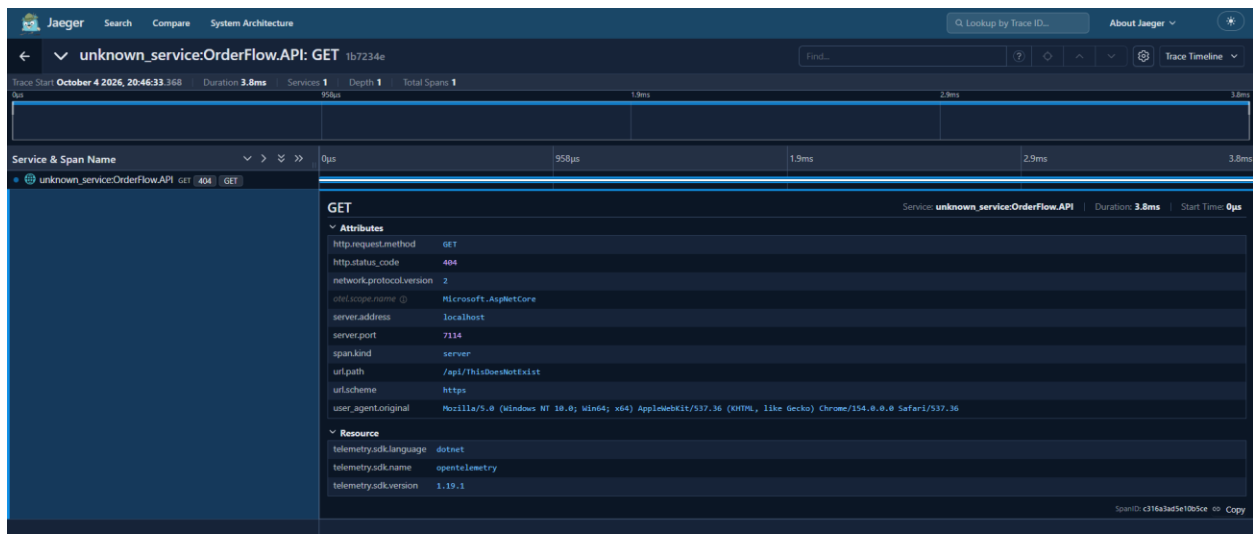
Distributed tracing shows the execution time and details of important OrderFlow operations.





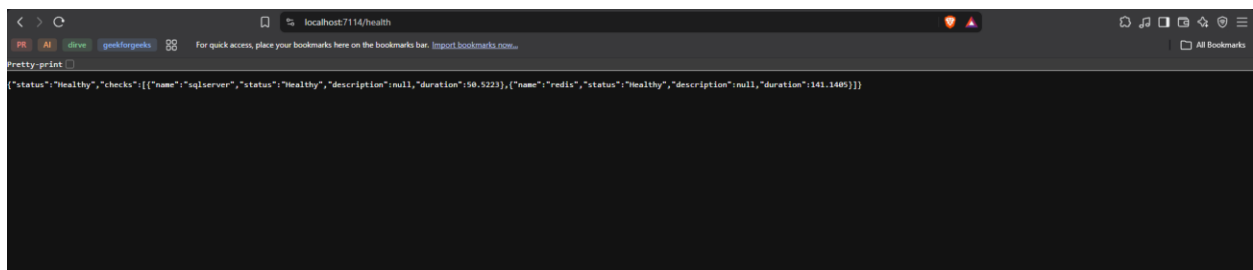
Tracing Failure Test

A GET request was sent to a non-existent endpoint (/api/ThisDoesNotExist), resulting in a 404 Not Found response, and the request was successfully captured and traced in Jaeger using OpenTelemetry.

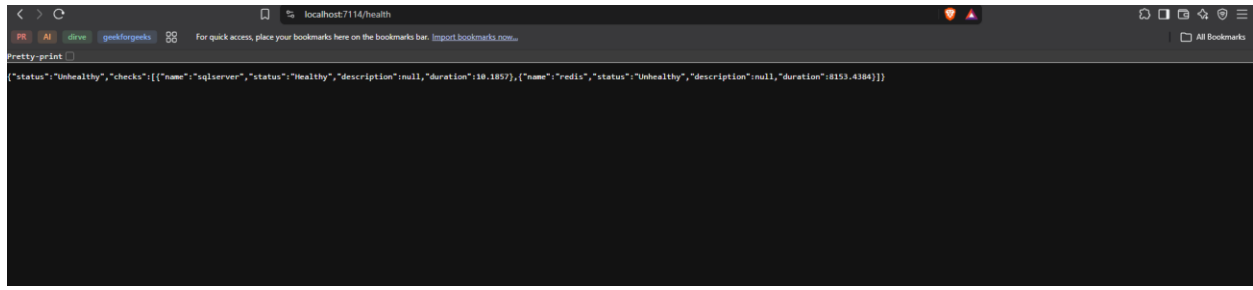


Health Check

Health checks verify that the API, SQL Server, and Redis dependencies are available.



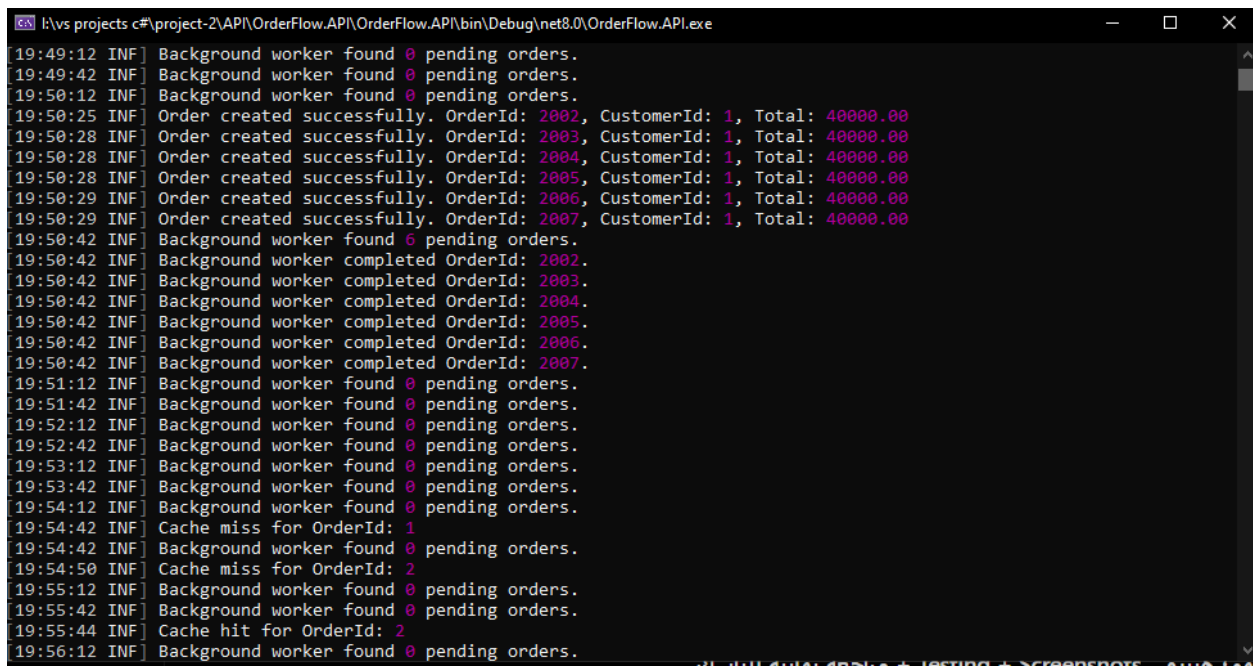
Redis Off



```
localhost:7114/health
Pretty-print
{"status":"Unhealthy","checks":[{"name":"sqlserver","status":"Healthy","description":null,"duration":18.1857}, {"name":"redis","status":"Unhealthy","description":null,"duration":8153.4384}]}
```

Logging

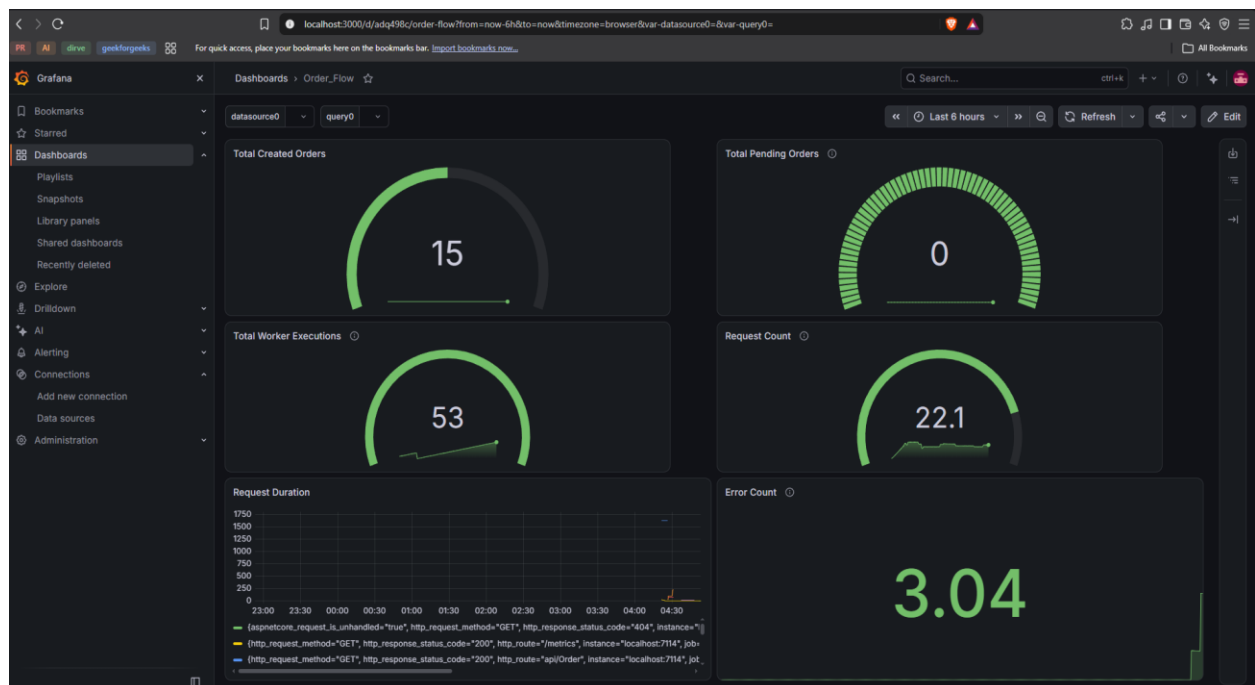
Structured logs provide useful information about order creation and background worker processing.



```
I:\vs projects c#\project-2\API\OrderFlow.API\OrderFlow.API\bin\Debug\net8.0\OrderFlow.API.exe
[19:49:12 INF] Background worker found 0 pending orders.
[19:49:42 INF] Background worker found 0 pending orders.
[19:50:12 INF] Background worker found 0 pending orders.
[19:50:25 INF] Order created successfully. OrderId: 2002, CustomerId: 1, Total: 40000.00
[19:50:28 INF] Order created successfully. OrderId: 2003, CustomerId: 1, Total: 40000.00
[19:50:28 INF] Order created successfully. OrderId: 2004, CustomerId: 1, Total: 40000.00
[19:50:28 INF] Order created successfully. OrderId: 2005, CustomerId: 1, Total: 40000.00
[19:50:29 INF] Order created successfully. OrderId: 2006, CustomerId: 1, Total: 40000.00
[19:50:29 INF] Order created successfully. OrderId: 2007, CustomerId: 1, Total: 40000.00
[19:50:42 INF] Background worker found 6 pending orders.
[19:50:42 INF] Background worker completed OrderId: 2002.
[19:50:42 INF] Background worker completed OrderId: 2003.
[19:50:42 INF] Background worker completed OrderId: 2004.
[19:50:42 INF] Background worker completed OrderId: 2005.
[19:50:42 INF] Background worker completed OrderId: 2006.
[19:50:42 INF] Background worker completed OrderId: 2007.
[19:51:12 INF] Background worker found 0 pending orders.
[19:51:42 INF] Background worker found 0 pending orders.
[19:52:12 INF] Background worker found 0 pending orders.
[19:52:42 INF] Background worker found 0 pending orders.
[19:53:12 INF] Background worker found 0 pending orders.
[19:53:42 INF] Background worker found 0 pending orders.
[19:54:12 INF] Background worker found 0 pending orders.
[19:54:42 INF] Cache miss for OrderId: 1
[19:54:42 INF] Background worker found 0 pending orders.
[19:54:50 INF] Cache miss for OrderId: 2
[19:55:12 INF] Background worker found 0 pending orders.
[19:55:42 INF] Background worker found 0 pending orders.
[19:55:44 INF] Cache hit for OrderId: 2
[19:56:12 INF] Background worker found 0 pending orders.
```

Grafana Dashboard

Grafana provides a visual overview of request traffic, duration, errors, orders, and pending orders.



Alerts

notify when important operational conditions exceed defined thresholds.

